

Literature Survey

TriageGuard: An Adversarially Robust AI Assistant for Memory Forensics and Cryptographic Malware Triage

Sagnik Basu (240953528)

Arjun Mittal (240953440)

Introduction

This literature survey examines twenty works, all published from 2021 onward, spanning every pillar TriageGuard integrates. Papers 1 to 10 (2024 onward, journal-first with one review paper) cover AI-assisted memory forensics, entropy- and header-based ransomware detection, measured cryptographic benchmarks for symmetric and asymmetric ciphers (AES, DES, 3DES, RSA, ElGamal, ECC), RSA weak-key cryptanalysis, JWT/OAuth 2.0 token security, and LLM robustness against adversarial input. Papers 11 to 20 (2021 to 2024) add memory-based malware detection, LLMs in digital forensics, recovering ransomware and TLS keys from memory, identifying cryptographic primitives in binaries, ransomware’s use of hybrid encryption, fast shared-prime detection for RSA, and the prompt-injection attacks and defences our red-team track tests. Table 1 illustrates each paper’s domain, methodology, and key findings alongside its reported cryptographic parameters (key size, plaintext/ciphertext size, encryption type, timing, and performance) where applicable, its limitations, and how it relates to three specific pillars of TriageGuard: memory forensics, cryptographic stress analysis, and AI robustness.

Table 1: Literature survey combining qualitative relevance analysis with reported cryptographic benchmark parameters, comparing twenty works against TriageGuard.

#	Author, Paper & Venue	Yr.	Domain Focus & Methodology	Key Findings & Observation	Key Size	PL Size	CT Size / Output	Encryption	Time Consuming	Performance	Limitation / Gap in Paper	Relevance: Memory Forensics	Relevance: Cryptographic Stress	Relevance: AI Robustness & Gap Addressed by TriageGuard
1	Baig, Ul Haq, Habib, “A Comparative Analysis of AES, RSA, and 3DES Encryption Standards based on Speed and Performance” (Management Science Advances)	2024	Symmetric vs. asymmetric cipher overview; literature-based comparative review of AES, DES, 3DES, RSA covering computation time and memory usage	Symmetric ciphers process faster than RSA; RSA offers stronger key-exchange properties. A review rather than a fresh experimental benchmark; no primary data of its own	AES 128/192/256-bit; DES 56-bit; 3DES 112-bit	Not experimentally varied	N/A	AES, DES, 3DES, RSA	Cites prior benchmarks showing AES/DES faster than RSA	Qualitative time/memory comparison only	Relies entirely on cited studies; no primary experimental data	None directly	Confirms the baseline speed hierarchy (AES > DES > RSA) that TriageGuard’s algorithm-classification module assumes	None directly
2	Banu, Rath, Gountia, “Analyzing cryptographic algorithm efficiency within graph-based encryption models” (Frontiers in Computer Science)	2025	Asymmetric cipher performance on real files; implements RSA/ElGamal in C#, benchmarks on text/image/audio files of 22 KB to 5,120 KB against AES-256/ECC baselines	RSA is faster but memory-heavier; ElGamal is slower but more memory-efficient; both far slower than AES-256. Real measured trade-off; single-machine benchmark	RSA/ElGamal/ECC 2048-bit; AES-256; 64/128-bit in early trials	22 KB to 5,120 KB (text/image/audio)	RSA ~169.82 KB memory for 22 KB input; ElGamal ~0.17 KB	RSA, ElGamal vs. AES-256, ECC	RSA 0.108 s vs. ElGamal 1.55 s for 22 KB text; AES-256 ~0.001 s	RSA throughput ~203 KB/s vs. ElGamal ~14 KB/s	Single-machine benchmark; no malware-context evaluation	File-level, not memory-resident, but technique is reusable	Very high; supplies real operating ranges for TriageGuard’s crypto-detection thresholds	None directly
3	Ruzai, Ying, Muhammad, Asbulah, Ariffin, “Concurrent factorization of RSA moduli via weak key equations” (AIMS Mathematics)	2024	RSA cryptanalysis; applies Diofantine approximation and Copersmith’s lattice technique to sets of RSA public keys	Polynomial-time factorization achievable when small-integer relationships hold among RSA parameters. Purely theoretical; no implementation	RSA moduli, structurally related, unspecified bit-lengths	N/A	N/A	RSA	Polynomial-time (theoretical)	Demonstrates concurrent factorization of multiple weak moduli	No automated tooling to detect these conditions in real malware samples	None directly	Very high; underlies TriageGuard’s RSA weak-key checks (small exponent, shared modulus)	None directly
4	Raghu, “A Context-Aware Framework for Secure Fintech APIs: Leveraging Java Spring Boot and OAuth 2.0 with Dynamic Token Adaptation” (J. Info. Systems Eng. & Mgmt.)	2025	OAuth 2.0/JWT adaptive security for fintech APIs; Spring Boot + Redis-based dynamic risk scoring adjusting JWT scope/TTL in real time, load-tested with JMeter	Adaptive tokens cut unauthorized access by 84% at a small latency cost. Security gains carry an explicit, quantified performance cost	RSA 2048-bit (JWT via RS256)	N/A (tokens, not files)	N/A; JWT byte-size not reported	OAuth 2.0/JWT (RS256) with dynamic scope/TTL	+3 ms median latency (10 ms vs. 7 ms baseline)	≥230 TPS vs. 250 baseline; 84% fewer unauthorized attempts	Tested on one GKE deployment; not evaluated against forged tokens embedded in malware	None directly	High; benchmark for acceptable JWT/OAuth hardening overhead	None directly
5	Rujichaikul, Rassameeroj, “Token-Based Authentication Monitoring System” (J. Cyber Security and Mobility)	2025	JWT/OAuth attack detection; 25 rule-based detection signatures tested via 70 test cases on a token-based web app	81.4% detection accuracy, 92% recall, low operational overhead. Prioritises recall over false-positive rate	N/A; monitors usage patterns, not the signing key	N/A	N/A	JWT access tokens + OAuth 2.0 refresh tokens (monitored)	9.7 ms avg. detection latency	81.4% accuracy; 2.3% CPU overhead; +18 MB memory	Struggles with VPN/IP-spoofing edge cases; no ML component	None directly	High; precedent for detection-accuracy/overhead trade-offs	None directly
6	Oh, Kim, Kim, Kim, “volGPT: Evaluation on triaging ransomware process in memory forensics with large language model” (Forensic Sci. Int.: Digital Investigation)	2024	AI-assisted memory forensics for ransomware; extracts process artefacts from memory images, feeds them to an LLM to triage ransomware-related processes	LLM-assisted triage improves identification of ransomware-relevant processes over raw plugin output. Prompt design and hallucination remain open issues	N/A	N/A	N/A	N/A (detects ransomware presence, not cipher parameters)	Not reported	Improved triage accuracy vs. plugin baseline	Never tests whether a crafted sample can bias the LLM’s own verdict	Very high; near-identical Stage 1 approach TriageGuard builds on	Low; detects ransomware but not encryption strength	Gap TriageGuard’s red-teaming track directly addresses
7	Sharma, Ghawaly, McCleary, Webb, Baggili, “ForensicLLM: A local large language model for digital forensics” (Forensic Sci. Int.: Digital Investigation)	2025	Locally-hosted LLM for forensic reports; fine-tunes a local LLM on forensic artefact data for investigator queries without cloud exposure	Local deployment matches cloud LLM Q&A performance with less data-exposure risk. Focuses on deployment/privacy, not adversarial evidence	N/A	N/A	N/A	N/A	Not reported	Comparable to cloud LLM performance	No test of manipulated input	High; validates TriageGuard’s own LLM-report design choice	None directly	Adds the missing layer TriageGuard tests: resistance to manipulated input
8	Casino, Hurley-Smith, Hernandez-Castro, Patsakis, “Not on my watch: ransomware detection through classification of high-entropy file segments” (Journal of Cybersecurity)	2025	Entropy-based ransomware/encryption detection; trains a classifier to distinguish genuinely encrypted segments from other high-entropy data (e.g. compressed files)	Outperforms naive entropy-threshold detectors, reduces false positives. Detects that data is encrypted, not which algorithm or how strong	N/A (detection, not a specific cipher benchmark)	File segments of varying entropy (size unspecified)	N/A	Detects encrypted vs. compressed high-entropy data (algorithm-agnostic)	Not reported	Improved false-positive rate over threshold baseline	No algorithm identification or key-strength evaluation	Moderate; file-level, technique reusable	Very high; direct basis for TriageGuard’s entropy module	None; no AI reporting layer
9	Gokcimen, “A novel system for strengthening security in large language models” (Alexandria Engineering Journal)	2025	LLM security hardening against manipulation; proposes an architectural defence layer for LLM deployments, evaluated against injection-style attacks	Measurable improvement in resistance to malicious input manipulation. General-purpose framework, not forensics-specific	N/A	N/A	N/A	N/A	Not reported	Reduced attack success rate (framework-level)	Not evaluated in a forensics/malware-triage context or against evidence-embedded attacks	None directly	None directly	Very high; parallels TriageGuard’s own attack-then-harden methodology
10	Barnes, Ghafarian, “Machine Learning-Based Static Ransomware Detection Using PE Header Features and SHAP Interpretation” (J. Cybersecurity and Privacy)	2026	Static PE-header-based ransomware classification; trains Random Forest, SVM, XGBoost on PE header features across three datasets; uses SHAP to interpret decisions	Strong classification accuracy; SHAP identifies the most indicative header fields. Static features are brittle against packing/obfuscation	N/A	N/A	N/A	N/A (static feature classification, not cipher benchmark)	Not reported	High classification accuracy (exact figures not in abstract)	Doesn’t test evasion or combine with entropy/behavioural signals	High; reinforces TriageGuard’s static-analysis stage	Moderate; complements entropy/key-level analysis	None; no adversarial-robustness component
11	Carrier, Victor, Tekeoglu, Lashkari, “Detecting Obfuscated Malware using Memory Feature Engineering” (ICISSP)	2022	ML malware detection from memory; extends the VolMemLyzer feature extractor (built on Volatility) and trains a stacked ensemble on the new MalMemAnalysis-2022 dataset	Memory features detect obfuscated malware with 99.00% accuracy and 99.02% F1-score	N/A	N/A	N/A	N/A	Not reported here	99.00% accuracy, 99.02% F1	Binary malicious/benign verdict; no explanation, no cryptographic analysis	Very high; confirms Volatility features carry the evidence TriageGuard’s Stage 1 extracts	None directly	None; no LLM layer. TriageGuard explains findings in a cited report instead of a label
12	Scanlon, Breitingher, Hargreaves, Hilgert, Sheppard, “ChatGPT for digital forensic investigation: The good, the bad, and the unknown” (Forensic Sci. Int.: Digital Investigation)	2023	LLMs in forensics; tests GPT-4 on artefact analysis, evidence search, script writing, anomaly detection, incident response and training scenarios	Some low-risk uses work, but many are unsuitable today: uploading evidence raises privacy issues and the user must catch the model’s mistakes	N/A	N/A	N/A	N/A	Not reported	Qualitative, per scenario	No systematic defence against wrong or manipulated output	High; motivates feeding the LLM findings, not raw evidence	None directly	Very high; TriageGuard adds mandatory citations and a rule cross-check so mistakes are caught automatically
13	Fernandez de Loaysa Babiano, Macfarlane, Davies, “Evaluation of live forensic techniques, towards Salsa20-based cryptographic ransomware mitigation” (Forensic Sci. Int.: Digital Investigation)	2023	Live memory forensics against ransomware; extracts Salsa20 keys and nonces from volatile memory while ransomware runs	Over 90% of Salsa20 key and nonce pairs recovered from memory, enough to decrypt files without paying, even with a unique key per file	Salsa20 256-bit key + 64-bit nonce	Victim files	Encrypted files (stream cipher, same length)	Salsa20 inside hybrid encryption	Not reported here	>90% key/nonce pairs recovered	Targets Salsa20 specifically; needs capture while the ransomware is running	Very high; shows modern ransomware keys live in RAM	Very high; supports TriageGuard’s ChaCha/Salsa20 detection	None

#	Author, Paper & Venue	Yr.	Domain Focus & Methodology	Key Findings & Observation	Key Size	PL Size	CT Size / Output	Encryption	Time Consuming	Performance	Limitation / Gap in Paper	Relevance: Memory Forensics	Relevance: Cryptographic Stress	Relevance: AI Robustness & Gap Addressed by TriageGuard
14	Baier, Basse, Hilgert, Lambertz, “TLS key material identification and extraction in memory: Current state and future challenges” (DFRWS USA)	2024	Systematisation of knowledge; reviews academic and in-the-wild methods for finding and extracting TLS key material from memory	Memory extraction of TLS keys is essential for decrypting malware traffic; lists open research challenges	TLS session keys and secrets	N/A	N/A	TLS (symmetric session keys)	Not reported	Qualitative survey	Survey; does not cover RSA weak-key testing or reporting	Very high; memory as a source of key material	High; links TriageGuard’s memory and pcap stages	None
15	Meijer, Moonsamy, Wetzels, “Where’s Crypto?: Automated Identification and Classification of Proprietary Cryptographic Primitives in Binary Code” (USENIX Security)	2021	Crypto identification in binaries; data-flow-graph isomorphism plus symbolic execution, released as an IDA plugin	Identifies and classifies both known and proprietary primitives, tested on public and proprietary binaries	Varies per primitive	N/A	N/A	Known and proprietary primitives	Not reported here	Not reported here	Per-binary analysis in a disassembler; not on whole memory images	Low	Very high; covers custom ciphers that TriageGuard’s constant signatures would miss	None
16	Oz, Aris, Levi, Uluagac, “A Survey on Ransomware: Evolution, Taxonomy, and Defense Solutions” (ACM Computing Surveys)	2022	Survey of ransomware evolution, taxonomy, and detection and defence techniques across platforms	Organises ransomware by target, infection and encryption behaviour, and reviews defences	Varies (symmetric + asymmetric hybrids)	Victim files	Encrypted files	Hybrid encryption, varies by family	Not reported	Survey	Survey; no tool	Moderate	High; motivates identifying which cipher a sample really uses	None
17	Pelofske, “An Efficient All-to-All GCD Algorithm for Low Entropy RSA Key Factorization” (IACR ePrint 2024/699)	2024	RSA weak-key cryptanalysis; binary-tree batch GCD to find RSA moduli that share a prime because of low-entropy key generation	About 6x faster than the standard remainder-tree batch GCD, same asymptotic scaling	RSA moduli	N/A	Shared prime factors	RSA	About 6x speedup over remainder-tree batch GCD	Speedup grows with the number of shared factors	Algorithmic; applied to key collections, not memory	None directly	Very high; same shared-prime test TriageGuard runs on carved keys	None
18	Greshake, Abdelnabi, Mishra, Endres, Holz, Fritz, “Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection” (ACM AISec)	2023	Indirect prompt injection; instructions planted in data an LLM application retrieves, shown on real systems	Retrieved data can take over an LLM-integrated application	N/A	N/A	N/A	N/A	Not reported	Demonstrated on deployed applications	General applications; no forensic setting; no measured defences	Low	None directly	Very high; memory artefacts are exactly such retrieved data
19	Hines, Lopez, Hall, Zarfati, Zunger, Kiciman, “Defending Against Indirect Prompt Injection Attacks With Spotlighting” (arXiv:2403.14720)	2024	Defence; marks untrusted input by delimiting, datamarking or encoding so the model can tell data from instructions	Reduces attack success from over 50% to under 2% in their experiments	N/A	N/A	N/A	N/A	Not reported	ASR from >50% to <2%	Single input-side defence; no output validation	None directly	None directly	Very high; TriageGuard’s defence D2. combined with output checks
20	Liu, Jia, Geng, Jia, Gong, “Formalizing and Benchmarking Prompt Injection Attacks and Defenses” (USENIX Security)	2024	Framework formalising prompt injection; benchmark of 5 attacks and 10 defences on 10 LLMs and 7 tasks	Combined attacks are most effective; existing defences are not sufficient on their own	N/A	N/A	N/A	N/A	Not reported	Benchmark across 10 LLMs	General tasks; no evidence-citation check	None directly	None directly	Very high; supports layering several defences as TriageGuard does

This table illustrates that real, measured cryptographic parameters exist for symmetric ciphers (Paper 1), asymmetric ciphers under realistic file loads (Paper 2), weak RSA configurations (Papers 3 and 17), and JWT/OAuth token security under production-like conditions (Papers 4 and 5), and that keys and cipher implementations can be recovered from memory and binaries (Papers 13 to 15). AI-assisted memory forensics (Papers 6, 7, 11, 12), entropy/header-based ransomware detection (Papers 8, 10, 16), and LLM robustness against adversarial input (Papers 9, 18 to 20) are each addressed independently elsewhere. No single existing work reports all of these side by side or feeds them into one automated, evidence-cited triage report, and the LLM-based forensic tools (Papers 6, 7, 12) never test what happens when the evidence itself carries injected instructions, which Paper 18 shows is realistic. TriageGuard’s contribution is to operationalise these benchmarked thresholds, key sizes and detection techniques into a single pipeline that flags a sample’s actual cryptographic and authentication risk, rather than just its presence, while also red-teaming and hardening its own AI reporting layer.

References

1. M. H. M. Baig, H. B. Ul Haq, and W. Habib, “A Comparative Analysis of AES, RSA, and 3DES Encryption Standards based on Speed and Performance,” *Management Science Advances*, vol. 1, no. 1, pp. 20–30, 2024.
2. Y. Banu, B. K. Rath, and D. Gountia, “Analyzing cryptographic algorithm efficiency within graph-based encryption models,” *Frontiers in Computer Science*, vol. 7, art. 1630222, 2025.
3. W. N. A. Ruzai, Y. Ying, K. N. Muhammad, M. A. Asbullah, and M. R. K. Ariffin, “Concurrent factorization of RSA moduli via weak key equations,” *AIMS Mathematics*, vol. 9, no. 10, pp. 28211–28231, 2024.
4. A. Raghu, “A Context-Aware Framework for Secure Fintech APIs: Leveraging Java Spring Boot and OAuth 2.0 with Dynamic Token Adaptation,” *Journal of Information Systems Engineering and Management*, vol. 10, no. 49s, pp. 367–377, 2025.
5. P. Rujichaikul and I. Rassameeroj, “Token-Based Authentication Monitoring System,” *Journal of Cyber Security and Mobility*, vol. 14, no. 4, pp. 777–798, 2025.
6. D. B. Oh, D. Kim, D. Kim, and H. K. Kim, “volGPT: Evaluation on triaging ransomware process in memory forensics with large language model,” *Forensic Science International: Digital Investigation*, vol. 49, p. 301756, Jul. 2024.
7. B. Sharma, J. Ghawaly, K. McCleary, A. M. Webb, and I. Baggili, “ForensicLLM: A local large language model for digital forensics,” *Forensic Science International: Digital Investigation*, vol. 52, p. 301872, Mar. 2025.
8. F. Casino, D. Hurley-Smith, J. Hernandez-Castro, and C. Patsakis, “Not on my watch: ransomware detection through classification of high-entropy file segments,” *Journal of Cybersecurity*, vol. 11, no. 1, p. tyaf009, 2025.
9. T. Gokcimen, “A novel system for strengthening security in large language models,” *Alexandria Engineering Journal*, vol. 75, pp. 100–112, Jan. 2025.
10. G. Barnes and A. Ghafarian, “Machine Learning-Based Static Ransomware Detection Using PE Header Features and SHAP Interpretation,” *Journal of Cybersecurity and Privacy*, vol. 6, no. 2, p. 58, 2026.
11. T. Carrier, P. Victor, A. Tekeoglu, and A. H. Lashkari, “Detecting Obfuscated Malware using Memory Feature Engineering,” in *Proc. 8th ICISSP*, 2022, pp. 177–188.
12. M. Scanlon, F. Bretinger, C. Hargreaves, J.-N. Hilgert, and J. Sheppard, “ChatGPT for digital forensic investigation: The good, the bad, and the unknown,” *Forensic Science International: Digital Investigation*, vol. 46, p. 301609, 2023.
13. L. Fernandez de Loaysa Babiano, R. Macfarlane, and S. R. Davies, “Evaluation of live forensic techniques, towards Salsa20-based cryptographic ransomware mitigation,” *Forensic Science International: Digital Investigation*, vol. 46, p. 301572, 2023.
14. D. Baier, A. Basse, J.-N. Hilgert, and M. Lambertz, “TLS key material identification and extraction in memory: Current state and future challenges,” in *Proc. DFRWS USA*, 2024.
15. C. Meijer, V. Moonsamy, and J. Wetzels, “Where’s Crypto?: Automated Identification and Classification of Proprietary Cryptographic Primitives in Binary Code,” in *Proc. 30th USENIX Security Symposium*, 2021.
16. H. Oz, A. Aris, A. Levi, and A. S. Uluagac, “A Survey on Ransomware: Evolution, Taxonomy, and Defense Solutions,” *ACM Computing Surveys*, vol. 54, no. 11s, 2022.
17. E. Pelofske, “An Efficient All-to-All GCD Algorithm for Low Entropy RSA Key Factorization,” Cryptology ePrint Archive, Paper 2024/699, 2024.
18. K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, “Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection,” in *Proc. 16th ACM AISec*, 2023.
19. K. Hines, G. Lopez, M. Hall, F. Zarfati, Y. Zunger, and E. Kiciman, “Defending Against Indirect Prompt Injection Attacks With Spotlighting,” arXiv:2403.14720, 2024.
20. Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, “Formalizing and Benchmarking Prompt Injection Attacks and Defenses,” in *Proc. 33rd USENIX Security Symposium*, 2024.